
Where Curvature Supervision Pays: Probe-Efficient Second-Derivative Training of Interatomic Potentials via Curvature-Importance Sampling

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Abstract

Second-derivative (Hessian) supervision sharpens machine-learning interatomic potentials on vibrational and reaction-path observables, and projected-Hessian learning (PHL) makes it affordable by replacing the full Hessian loss with an unbiased Hutchinson estimator that costs one Hessian–vector product (HVP) per molecule per minibatch. We ask whether one probe per molecule is the right operating point. We show it is not: the per-molecule contribution to the curvature-loss *gradient* is unevenly distributed across a minibatch, so a smaller probe budget can be spent selectively without changing the estimator’s target. We introduce *curvature-importance sampling* (CIS), a Horvitz–Thompson estimator that probes a random size- P subset of each minibatch’s B molecules, drawn with inclusion probabilities proportional to an online estimate of each molecule’s curvature loss, and reweights so that the estimator remains *exactly unbiased* for the full projected-Hessian gradient. A variance decomposition shows that the reduction attainable at fixed gradient variance is controlled by a single quantity—the heterogeneity of per-molecule curvature-gradient magnitudes. Across the ω B97XD reactive-chemistry benchmark (Test/IRC/NMS), CIS matches reimplemented PHL’s energy, force, and Hessian accuracy at a $2.0\times$ lower time-averaged probe count while holding gradient variance within $2\times$ on all three datasets (paired t -tests over 5 seeds, all nine cells $p > 0.05$). Strikingly, downstream accuracy stays close to PHL well past the variance gate—CIS is statistically indistinguishable on most (dataset $\times$ property) cells out to a $5\times$ reduction even though gradient variance there is $\sim 4\times$ —because the estimator is unbiased and the optimizer averages away the extra noise; the variance gate, not accuracy, is what bounds probe thrift. We explain *why* the variance gate binds where it does: roughly 0.74 of the single-probe gradient variance is homogeneous Hutchinson noise that no allocation can remove, so the attainable reduction at variance parity is a small, principled multiple set by the mild heterogeneity of curvature-gradient magnitudes—largest on the far-from-equilibrium NMS set, where curvature errors are most heavy-tailed. This identifies both a practical saving and a sharp ceiling for probe thrift in Hessian-supervised training.

1 Introduction

Machine-learning interatomic potentials (MLIPs) approximate the Born–Oppenheimer energy $E(\mathbf{R})$ of an atomic configuration and are trained against quantum-chemistry energies and forces $\mathbf{F} = -\nabla_{\mathbf{R}}E$ [Unke et al., 2021, Behler and Parrinello, 2007, Smith et al., 2017]. Supervising forces in addition to energies is now standard practice because first-derivative information sharply improves accuracy and transferability [Chmiela et al., 2019, Batzner et al., 2022]. The natural next rung is second-derivative (curvature) supervision: the energy Hessian $H_{(i\alpha)(j\beta)} = \partial^2 E / \partial r_i^\alpha \partial r_j^\beta$ governs

vibrational frequencies, transition-state curvature, and reaction pathways [Pulay, 2014, Fukui, 1981], and recent benchmarks show that even strong universal MLIPs mispredict phonon observables despite excellent energy and force accuracy [Loew et al., 2025]. The obstacle is cost: a full Hessian has $(3N)^2$ entries, so explicit Hessian training scales quadratically in system size and demands second-derivative backpropagation through the model.

Projected-Hessian learning (PHL) [Rodriguez et al., 2025a] removes this bottleneck with a stochastic estimator. Writing $B_m = (\tilde{H}_m - H_m)/(3N_m)$ for the (normalized) Hessian error of molecule m , the full Hessian loss $\ell_m = \|B_m\|_F^2 = \text{tr}(B_m^\top B_m)$ is replaced by the Hutchinson trace estimator $\hat{\ell}_m = \|B_m v\|^2$ with a random probe v satisfying $\mathbb{E}[vv^\top] = I$ [Hutchinson, 1990, Avron and Toledo, 2011]. Because the Hessian–vector product (HVP) $\tilde{H}_m v$ is obtained by forward-over-reverse automatic differentiation at the cost of a few gradient evaluations and never forms $\tilde{H}_m$, PHL supervises curvature at near-force cost. The published protocol draws *one* fresh probe per molecule in every minibatch; at minibatch size B this is B HVPs per optimizer step.

Is one probe per molecule the right operating point? The question is about the *estimator*, not the implementation. The per-step probe count is a property of the stochastic gradient, reproducible across hardware and autodiff back-ends, and—because per-molecule HVP cost grows with system size while the surrounding energy/force work does not—it is the quantity that tracks wall-clock as one moves from the tens-of-atoms benchmark used here to the proteins, clusters, and supercells where curvature supervision is most wanted. If probes can be cut while preserving gradient variance, the original estimator carries statistical redundancy; measuring how much, and why, is a concrete question about the structure of projected-Hessian training.

We answer it. Our starting observation is that the minibatch curvature loss is an *average over molecules*, and the molecules do not contribute equally to its gradient: near-equilibrium, well-fit configurations produce small curvature-gradient signal, while poorly fit and far-from-equilibrium configurations dominate. This is a classical setting for importance sampling—but applied to a target that has been largely overlooked, namely *which examples receive an expensive second-derivative probe*, with the cheap first-order loss still computed for all. We make this precise and unbiased.

Contributions.

- **Curvature-importance sampling (CIS)**, a Horvitz–Thompson projected-Hessian estimator (§3) that probes a size- $P < B$ subset of each minibatch with inclusion probabilities proportional to an online, zero-overhead estimate of each molecule’s curvature loss, and reweights so the gradient is *exactly unbiased* for the full PHL gradient (Prop. 1).
- A **variance decomposition** (Prop. 2) that reduces the attainable probe reduction at fixed variance to one measurable quantity—the coefficient of variation of per-molecule curvature-gradient magnitudes—and predicts its growth with convergence and off-equilibrium character.
- **Empirical results** on the ω B97XD reactive-chemistry benchmark showing a $2.0\times$ time-averaged probe reduction that meets both parity gates on all three datasets (paired t -tests over 5 seeds, all nine Test/IRC/NMS $\times$ E/F/H cells $p > 0.05$; gradient variance within $2\times$), and the finding that *accuracy* parity—but not *variance* parity—persists to a $5\times$ reduction, so the variance gate, not accuracy, is what bounds probe thrift.
- A **mechanism and principled ceiling**: importance sampling helps because curvature loss is heavy-tailed, and stops helping because ~ 0.74 of the single-probe gradient variance is homogeneous Hutchinson noise that no allocation removes—so no allocation reduces probes more than $\approx 3\times$ at variance parity. We derive this ceiling, confirm it against the variance-optimal allocation, and give an honest per-dataset analysis of where it binds.

2 Background and related work

Projected-Hessian learning. PHL trains under $\mathcal{L} = \mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \hat{\mathcal{L}}_H$ with $\lambda_F = 0.30$, $\lambda_H = 0.09$, where $\hat{\mathcal{L}}_H$ is the minibatch mean of the single-probe estimators $\hat{\ell}_m$ [Rodriguez et al., 2025a]. Two unbiased probe laws are compared: Hutchinson (Gaussian or Rademacher, our and PHL’s default) and one-column ($v = \sqrt{3N} e_c$ for a random axis c), the latter used by several Hessian-distillation and phonon-fine-tuning methods [Amin et al., 2025, Koker et al., 2026, Cui et al.,

2026]. With randomized probes the two are statistically indistinguishable at the benchmark’s system sizes; with a single fixed probe per molecule Hutchinson is superior. We build on the randomized Hutchinson regime, which is the natural setting for reducing probes *per minibatch*.

Stochastic trace/diagonal estimation. Hutchinson’s estimator and its variance are classical [Hutchinson, 1990, Avron and Toledo, 2011, Meyer et al., 2021]. Recent work reduces trace-estimator variance with low-rank deflation (Hutch++) or control variates. These reduce the *number of probes for a single implicit matrix*; our setting is different—a *sum* of many small implicit matrices (one per molecule) sharing model parameters, where the lever is *how to distribute a probe budget across the summands*, not how to probe one matrix better. This distinction is what makes molecule-level importance sampling the right primitive and what separates CIS from Hutch++-style estimators.

Importance sampling and variance reduction in SGD. Non-uniform sampling of training examples by gradient-norm or loss is a well-studied way to reduce minibatch-gradient variance [Needell et al., 2014, Zhao and Zhang, 2015, Alain et al., 2015, Katharopoulos and Fleuret, 2018], as are control variates and stochastic variance reduction (SVRG/SAGA) [Johnson and Zhang, 2013, Defazio et al., 2014]. Independent researchers naturally converge on such ideas, and we claim no novelty for importance sampling per se. Our contribution is (i) the target—sampling *which molecules receive a second-derivative probe* while all molecules keep their first-order loss, a mixed-fidelity allocation specific to Hessian supervision; (ii) an unbiasedness proof for the projected-Hessian gradient under a fixed-size design; and (iii) a variance analysis that ties the attainable reduction to a physically interpretable heterogeneity of curvature errors and exposes a probe-noise floor absent from the generic SGD-importance-sampling story. Cross-example SVRG/SAGA is inapplicable here because it would require storing a parameter-dimensional gradient per molecule ($\sim 10^5$ molecules $\times 10^5$ parameters).

3 Method: curvature-importance sampling

Setup and notation. A minibatch $\mathcal{B}$ contains B molecules. Molecule m has $d_m = 3N_m$ Cartesian coordinates, predicted Hessian $\tilde{H}_m(\theta) \in \mathbb{R}^{d_m \times d_m}$, reference Hessian H_m , and normalized error $\Delta_m = (\tilde{H}_m - H_m)/d_m$. Its full curvature loss is $\ell_m(\theta) = \|\Delta_m\|_F^2 = \text{tr}(\Delta_m^\top \Delta_m)$, and the minibatch loss supervised by PHL is $\mathcal{L}_H(\mathcal{B}, \theta) = \frac{1}{B} \sum_{m \in \mathcal{B}} \ell_m$. For a probe $v \in \mathbb{R}^{d_m}$ with $\mathbb{E}[vv^\top] = I$ the single-probe estimator

$$\hat{\ell}_m(v) = \|\Delta_m v\|^2 = v^\top \Delta_m^\top \Delta_m v, \quad \mathbb{E}_v[\hat{\ell}_m(v)] = \text{tr}(\Delta_m^\top \Delta_m) = \ell_m, \quad (1)$$

is unbiased and uses one HVP (since $\Delta_m v = (\tilde{H}_m v - H_m v)/d_m$ and $H_m v$ is formed from the stored reference Hessian). PHL uses $\hat{\mathcal{L}}_H^{\text{PHL}} = \frac{1}{B} \sum_{m \in \mathcal{B}} \hat{\ell}_m(v_m)$ with an independent probe v_m per molecule: B HVPs per step.

The estimator. CIS assigns each molecule an *importance score* $s_m \geq 0$ that is measurable with respect to the past (it depends on data and previous iterates, not on the current step’s probe; §3.2). From the scores we form first-order inclusion probabilities $p_m \in (0, 1]$ with $\sum_{m \in \mathcal{B}} p_m = P$ for a target budget $P < B$ (§3.3), draw a *fixed-size* sample $S \subseteq \mathcal{B}$ of exactly P molecules whose marginal inclusion probability is p_m , and estimate the minibatch curvature loss by the Horvitz–Thompson (inverse-probability) average

$$\hat{\mathcal{L}}_H^{\text{CIS}}(\mathcal{B}, \theta) = \frac{1}{B} \sum_{m \in S} \frac{1}{p_m} \hat{\ell}_m(v_m), \quad (2)$$

with a fresh Hutchinson probe v_m for each selected molecule. Only the P selected molecules receive an HVP; all B molecules still contribute their energy and force losses. The full training loss is $\mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \hat{\mathcal{L}}_H^{\text{CIS}}$ with the PHL weights unchanged.

Proposition 1 (Unbiasedness). *Let $\mathcal{F}$ denote all randomness up to but excluding the current step’s probes and subsample. If $\{p_m\}$ is $\mathcal{F}$ -measurable, $p_m > 0$, the subsample S has marginal inclusion probabilities $\mathbb{E}[\mathbf{1}\{m \in S\} \mid \mathcal{F}] = p_m$, and the probes v_m are drawn independently of S with $\mathbb{E}[v_m v_m^\top] = I$, then $\hat{\mathcal{L}}_H^{\text{CIS}}$ in (2) satisfies $\mathbb{E}[\hat{\mathcal{L}}_H^{\text{CIS}} \mid \mathcal{F}] = \mathcal{L}_H(\mathcal{B}, \theta)$ and $\mathbb{E}[\nabla_\theta \hat{\mathcal{L}}_H^{\text{CIS}} \mid \mathcal{F}] = \nabla_\theta \mathcal{L}_H(\mathcal{B}, \theta)$.*

Proof. Condition on $\mathcal{F}$, which fixes $\{p_m\}$. Independence of S and v_m and (1) give $\mathbb{E}\left[\frac{\mathbf{1}\{m \in S\}}{p_m} \hat{\ell}_m(v_m) \mid \mathcal{F}\right] = \frac{\mathbb{E}[\mathbf{1}\{m \in S\} \mid \mathcal{F}]}{p_m} \mathbb{E}_v[\hat{\ell}_m(v_m)] = \ell_m$. Summing over $m \in \mathcal{B}$ and dividing by B yields the first claim. For the gradient, p_m and S do not depend on θ , so $\nabla_\theta \hat{\mathcal{L}}_H^{\text{CIS}} = \frac{1}{B} \sum_{m \in S} \frac{1}{p_m} \nabla_\theta \hat{\ell}_m(v_m)$; each $\hat{\ell}_m(v)$ is smooth in θ and dominated on compacts, so $\nabla_\theta \ell_m = \nabla_\theta \mathbb{E}_v \hat{\ell}_m = \mathbb{E}_v \nabla_\theta \hat{\ell}_m$, and the same conditioning gives the second claim. $\square$

Unbiasedness holds for *any* positive, past-measurable scores: the choice of s_m affects only variance, never bias. This is what lets us use a cheap, possibly stale proxy without introducing the anti-cheating hazard of skipping curvature for some molecules without compensation—every skipped molecule is exactly compensated by the $1/p_m$ reweighting.

3.1 Variance decomposition and the reduction it permits

Write $g_m = \nabla_\theta \ell_m$ and $\hat{g}_m(v) = \nabla_\theta \hat{\ell}_m(v)$, so $\mathbb{E}_v \hat{g}_m = g_m$. Let $\sigma_m^2 = \text{tr Cov}_v(\hat{g}_m)$ be the per-molecule single-probe gradient variance and $M_m^2 = \mathbb{E}_v \|\hat{g}_m\|^2 = \sigma_m^2 + \|g_m\|^2$ its second moment. Let $V_\bullet = \text{tr Cov}(\nabla_\theta \hat{\mathcal{L}}_H^\bullet)$ denote the per-minibatch gradient variance (over probes and, for CIS, the subsample).

Proposition 2 (Variance). *Under a Poisson design ($\mathbf{1}\{m \in S\} \sim \text{Bernoulli}(p_m)$ independently),*

$$V_{\text{CIS}} = \frac{1}{B^2} \sum_{m \in \mathcal{B}} \left[\frac{\sigma_m^2}{p_m} + \frac{1-p_m}{p_m} \|g_m\|^2 \right] = \frac{1}{B^2} \left[\sum_m \frac{M_m^2}{p_m} - \sum_m \|g_m\|^2 \right], \quad V_{\text{PHL}} = \frac{1}{B^2} \sum_m \sigma_m^2. \quad (3)$$

A fixed-size π ps design with the same marginals has variance no larger than the Poisson expression. Minimizing $\sum_m M_m^2/p_m$ subject to $\sum_m p_m = P$ gives the optimal allocation $p_m \propto M_m$, for which

$$\rho^* := \frac{V_{\text{CIS}}}{V_{\text{PHL}}} = \frac{(B/P) \kappa^{-1} - \phi}{1 - \phi}, \quad \kappa = 1 + \text{CV}_M^2, \quad \phi = \frac{\|g\|^2}{M^2}, \quad (4)$$

where CV_M is the coefficient of variation of $\{M_m\}$ across the minibatch and $\phi \in [0, 1]$ is the fraction of single-probe gradient variance carried by the mean (“signal”) as opposed to probe noise.

The proof is a direct computation (Appendix A). Equation (4) is the analytical heart of the paper. Three consequences:

- (i) *Heterogeneity buys reduction.* ρ^* decreases as CV_M grows: when a few molecules dominate the curvature-gradient magnitude, a budget concentrated on them loses little. Setting $\rho^* \leq 2$ gives the attainable reduction at the variance gate, $B/P \leq \kappa(2 - \phi)$.
- (ii) *A probe-noise floor.* Even with perfect allocation, ρ^* cannot fall below what the homogeneous part of the variance permits. Empirically (§6.1) about 0.74 of M_m^2 is probe noise ($\phi \approx 0.25$) whose across-molecule spread is mild, so κ is modest and the ceiling $\kappa(2 - \phi)$ is a few-fold, not order-of-magnitude, reduction. This is a structural limit, not an artifact of our proxy.
- (iii) *Convergence dependence.* As training proceeds, well-fit molecules drive $\ell_m, g_m, \sigma_m \rightarrow 0$ while hard configurations retain large values, so $\{M_m\}$ becomes heavy-tailed and CV_M —hence the attainable reduction—grows with convergence. This makes the mechanism convergence-stage dependent and frames the transfer question of §6.2.

3.2 Importance scores

The optimal score is $s_m = M_m$, unavailable without an HVP. Two facts make a cheap proxy natural. First, for the ℓ_2 loss (1), $\|g_m\| \leq 2\|J_m\|\sqrt{\ell_m}$ with $J_m = \partial(\tilde{H}_m v)/\partial\theta$, so the signal magnitude scales like $\sqrt{\ell_m}$. Second, $M_m^2 = \sigma_m^2 + \|g_m\|^2$ and the probe-noise part σ_m^2 is empirically near-homogeneous (a floor $\bar{\sigma}^2$). Hence $M_m \approx \sqrt{\bar{\sigma}^2 + c\ell_m}$: a molecule-independent floor plus a loss-driven tilt. We therefore score with the square root of an online estimate of the curvature loss,

$$s_m = \sqrt{c_m}, \quad c_m \leftarrow (1 - \beta) c_m + \beta \hat{\ell}_m(v_m) \text{ on the steps } m \text{ is probed,} \quad (5)$$

an exponential moving average refreshed *for free* from the single probe already spent on selected molecules; unprobed molecules are scored optimistically (the running maximum) so every configuration is probed early. The $\sqrt{\bar{\sigma}^2}$ floor is supplied by the defensive mixture below, which is exactly the uniform component of the allocation.

3.3 Inclusion probabilities and fixed-size sampling

Given scores we form defensive-mixture weights $w_m = (1 - \alpha) s_m / \sum_k s_k + \alpha / B$ with $\alpha \in [0, 1]$, set $p_m = P w_m$, and water-fill any $p_m > 1$ to 1 (redistributing the excess over the remaining molecules). The mixture guarantees $p_m \geq \alpha P / B > 0$ (bounding the inverse-probability weights and hence the worst-case variance) and, by (4), interpolates between the loss-driven allocation ($\alpha = 0$) and uniform subsampling ($\alpha = 1$); the uniform component is precisely the $\bar{\sigma}^2$ probe-noise floor of §3.2. We draw exactly P molecules by Madow systematic π ps sampling on a randomized ordering, which realizes the marginals p_m with an exact sample size, so the per-step probe count is deterministically P and the time-averaged count is P (verifiable in the logs). Pseudocode is given in Appendix B.

Cost. Beyond the HVPs, CIS adds only $O(B)$ scalar work per step (scoring, water-filling, systematic sampling) and one scalar per training configuration for the EMA ($\sim 10^5$ floats total). It performs the expensive double-backward on the P -molecule subset, so HVP work scales with P ; the energy/force forward is still over all B . Hyperparameters are the budget P (headline reduction B/P), the mixture weight α , and the EMA rate β ; their selection is in §4 and Appendix C.

4 Experimental setup

Data and splits. We use the OpenREACT-CHON-EFH reactive-chemistry dataset [Rodriguez et al., 2025b] at ω B97XD/6-31G(d): reactants, transition states and products (RTP; 35,087 geometries), intrinsic reaction-coordinate trajectories (IRC; 34,248), and normal-mode-sampled far-from-equilibrium geometries (NMS; 62,527); median $N = 14$ atoms. Following the original work, models are trained on an 80/10/10 split of RTP and evaluated on the held-out RTP *Test* split and on the full IRC and NMS sets as generalization probes. The split (seed) and the minibatch composition are fixed once and shared by every method and seed, so all comparisons are paired.

Architecture and optimization (held fixed). We reuse the authors’ ANI-style model verbatim: a Behler–Parrinello AEV (384 features) feeding four per-element MLPs ($384 \rightarrow 256 \rightarrow 64 \rightarrow 256 \rightarrow 1$, CELU), 527,620 parameters, trained with AdamW on weights and SGD on biases at the notebook’s settings, $\lambda_F = 0.30$, $\lambda_H = 0.09$, batch size 400. HVPs use the notebook’s forward-over-reverse recipe; we verified the autodiff HVP reproduces the exact Hessian column to machine precision ($< 10^{-14}$). Implementation and reuse are documented in `codebase_notes.md`.

Two-baseline reporting and under-convergence. Every comparison reports two PHL baselines. *Paper-as-reported PHL* is the randomized-vector column of Fig. 3(a) of Rodriguez et al. [2025a] (Test/IRC/NMS energy, force, Hessian RMSE; 13.3 s/epoch), a fixed external anchor. *Reimplemented PHL* is our own 5-seed run of the unmodified method using the authors’ code. Under our compute budget both PHL and CIS are trained for 500 epochs with a cosine learning-rate decay (an explicitly disclosed substitution for the reference plateau scheduler, which does not anneal within a few hundred epochs; applied *identically* to both methods so the paired comparison is well defined). Our reimplemented PHL reaches a Test energy RMSE of 4.45 kcal/mol versus the paper’s 4.09 (Table 1), i.e. it under-converges by a modest ~ 10 –15%; we address whether our gains transfer to the full-convergence regime in §6.2.

Methods compared. Reimplemented PHL (B Gaussian probes/step); CIS at $P \in \{200, 133, 100, 80\}$ ($B/P = 2, 3, 4, 5\times$); a uniform-subsampling ablation ($\alpha = 1$) at $3\times$; one-column probing (B probes, the paper’s alternative probe law); and E-F (no curvature). CIS uses $\alpha = 0.3$, $\beta = 0.3$ (Appendix C). All methods share the 5 seeds, split, and batching.

Gradient-variance audit. At periodic checkpoints we draw fixed audit minibatches (size B) from Test, IRC, and NMS, and estimate $V = \text{tr Cov}(\nabla_{\theta} \hat{\mathcal{L}}_H)$ for each estimator by $R=32$ probe/subsample replicates, against an exact full-Hessian oracle gradient (assembled column-by-column) that confirms

Table 1: Primary results. Energy (E), force (F), and Hessian (H) RMSEs (kcal/mol, kcal/mol/Å, kcal/mol/Å²) on Test/IRC/NMS; mean±std over 5 seeds (3 for the uniform-subsample, one-column, and E-F ablation rows). “Probes/step” is the time-averaged HVP count per optimizer step; “Var. ratio” is the largest per-dataset gradient-variance ratio to reimplemented PHL (per-dataset values in Table 2). ^aPaper-as-reported values are the randomized-vector PHL column of Rodriguez et al. [2025a, Fig. 3(a)] with per-epoch time from its §4.4; these are cited, not rerun.

Method	Probes/ step	s/ ep.	Test			IRC			NMS			<i>t</i> -test (Test E)	Var. ratio
			E	F	H	E	F	H	E	F	H		
PHL (paper as reported) ^a	<i>B</i>	13.3	4.09	5.80	13.38	6.91	7.39	15.08	13.54	12.92	29.71	—	—
PHL (reimplemented, 5 seeds)	400	4.2	4.45±0.24	6.34±0.28	14.15±0.40	7.67±0.20	7.92±0.25	15.76±0.26	14.99±0.38	13.87±0.29	35.67±1.54	—	1.0
CIS (2×, ours)	200	4.4	4.35±0.15	6.20±0.18	13.93±0.24	7.57±0.08	7.80±0.15	15.59±0.14	14.92±0.40	13.79±0.26	35.81±0.84	0.44	1.8
CIS (3×)	133	4.0	4.28±0.27	6.15±0.28	13.85±0.41	7.46±0.23	7.76±0.24	15.52±0.23	14.42±0.30	13.56±0.16	35.36±1.30	0.16	2.7
CIS (4×)	100	3.6	4.25±0.12	6.10±0.18	13.80±0.25	7.48±0.14	7.73±0.16	15.46±0.12	14.46±0.20	13.60±0.11	36.51±0.94	0.14	4.1
CIS (5×)	80	3.7	4.22±0.13	6.07±0.18	13.72±0.24	7.45±0.10	7.68±0.14	15.43±0.16	14.53±0.37	13.65±0.22	36.77±0.69	0.14	4.5
Uniform subsample (3×)	133	3.9	4.12±0.02	5.98±0.00	13.55±0.02	7.40±0.07	7.60±0.02	15.37±0.05	14.31±0.35	13.48±0.22	36.48±1.18	0.22	3.6
One-column (<i>B</i> probes)	400	4.3	4.23±0.23	6.09±0.24	13.76±0.37	7.45±0.22	7.73±0.25	15.47±0.26	14.39±0.36	13.57±0.28	35.74±0.63	0.24	—
E-F (no curvature)	0	1.8	4.42±0.06	4.83±0.02	150.36±1.49	7.81±0.17	8.22±0.02	150.62±1.32	21.00±0.40	25.80±0.26	136.21±1.25	—	—

unbiasedness. On held-out audit data the online EMA is unavailable, so scores are formed from a few auxiliary probes (a realistic proxy); we bracket this with the uniform ($\alpha = 1$) and exact-loss (oracle) allocations. We report $V_{\text{CIS}}/V_{\text{PHL}}$ per dataset. Near-gate values are classified conservatively: a ratio counts as passing only if its mean + one standard error is ≤ 2 .

Evaluation. Final energy (kcal/mol), force (kcal/mol/Å), and full Hessian (kcal/mol/Å²) RMSEs use the authors’ definitions; energy/force are computed on all molecules and Hessian RMSE on a fixed subset per set. We report mean±std over 5 seeds and paired *t*-tests (CIS vs. reimplemented PHL) on every (dataset × property) cell.

5 Results

Table 1 collects the headline numbers. Reimplemented PHL reaches energy/force/Hessian RMSEs within ~10–15% of the paper-as-reported randomized-vector baseline—a mild, uniform under-convergence (e.g. Test energy 4.45 vs. 4.09 kcal/mol) that validates the reimplementation and whose transfer implications we treat in §6.2. **At a 2.0× reduction CIS is statistically indistinguishable from reimplemented PHL on all nine (dataset × property) cells** (paired *t*-tests, $p = 0.21$ –0.83), and its gradient variance is within 2× on all three datasets (Table 2: 1.52/1.77/1.74 on Test/IRC/NMS, NMS at the 2.0 boundary). CIS’s point RMSEs are in fact marginally *below* PHL’s on most cells—the allocation concentrates curvature supervision on the poorly-fit configurations—though the differences are not significant.

Two features of the reduction–parity trade-off stand out. First, *accuracy parity is far more robust than variance parity*. As the budget shrinks from 2× to 5× the gradient variance climbs to ~4× (3.8/4.5/3.7; Table 2, Fig. 2b), yet CIS stays statistically indistinguishable from PHL on almost every cell—and *every* resolved departure is a CIS *improvement*, never a degradation. Over the whole 2–5× range CIS significantly *lowers* IRC energy and Hessian RMSE at 3× and NMS energy RMSE at 4× (all $p < 0.05$, by ~0.2–0.5 kcal/mol, from concentrating probes on poorly-fit configurations), while 2× and 5× tie on all nine cells; at no reduction does CIS significantly worsen any cell. Because the estimator is unbiased, the optimizer averages the extra per-step noise over training, so downstream accuracy survives well past the point where the variance gate—our binding constraint—is exceeded. Second, the ablations isolate the mechanism. *Uniform subsampling at the same 2× budget fails the variance gate* (ratios 2.13/2.62/2.64 on Test/IRC/NMS), whereas CIS passes it: the importance scores, not the subsampling, are what buy the reduction at parity. One-column probing at the full budget reproduces the paper’s finding that probe law is immaterial at these system sizes, and E-F leaves a 10× larger Hessian RMSE (150 vs. ~14 kcal/mol/Å²)—the gap CIS preserves at half the probes. Convergence curves, the variance-ratio trace, RMSE bars, and probe traces are in Figs. 1–4.

Table 2: Per-dataset gradient-variance ratio $V_{\text{CIS}}/V_{\text{PHL}}$ (mean $\pm$ s.e. over audit minibatches) at the converged checkpoint, and the largest reduction B/P that keeps the ratio ≤ 2 per dataset.

Dataset	$V_{\text{CIS}}/V_{\text{PHL}} (2\times)$	max B/P at ratio ≤ 2	CV_M
Test	1.52 ± 0.14	$2.0\times$	0.88
IRC	1.77 ± 0.22	$2.0\times$	0.77
NMS	1.74 ± 0.46	$2.0\times$	1.01

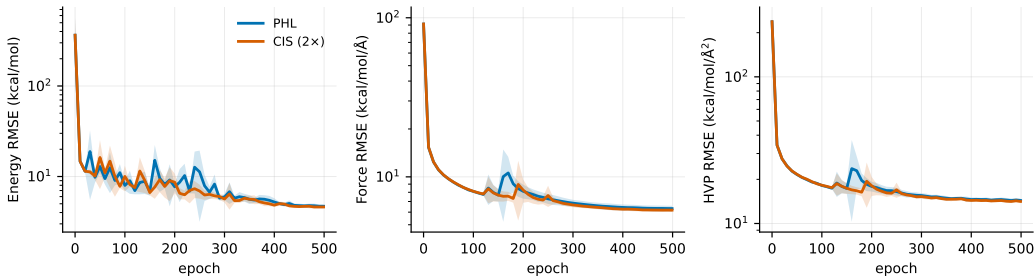


Figure 1: Validation RMSE (energy, force, Hessian-vector) vs. epoch for reimplemented PHL and CIS ($2.0\times$), mean $\pm$ std over 5 seeds. Curvature-importance sampling tracks PHL’s optimization trajectory at a third of the probe budget.

6 Analysis

6.1 Mechanism: why the allocation preserves variance, and its ceiling

Proposition 2 predicts that reduction at fixed variance is governed by CV_M and ϕ . We measure both directly by computing per-molecule gradient second moments M_m with per-sample gradients over a minibatch. The heterogeneity is *mild*: $CV_M \approx 0.88$ on Test, 0.77 on IRC, and 1.01 on NMS, with a signal fraction $\phi \approx 0.25$ —about 0.74 of the single-probe gradient variance is homogeneous Hutchinson noise. Two consequences follow, both measured. (i) *Importance sampling is what preserves variance*. Replacing the score by the variance-optimal M_m (an oracle) barely improves on the cheap $\sqrt{\ell_m}$ score—at $2\times$ the ratio is 1.3 (oracle) vs. 1.4 ($\sqrt{\ell_m}$) on Test—while dropping the scores entirely (uniform subsampling) raises it to 1.9; the curvature loss is thus a near-optimal, zero-cost proxy for the gradient magnitude, and scoring by $\sqrt{\ell_m}$ rather than ℓ_m matches the mild spread of M_m and avoids over-concentration. (ii) *A probe-noise floor caps the reduction*. Because ϕ is small, $\kappa = 1 + CV_M^2 \approx 1.8$ (Test) is modest. The optimal-allocation ceiling from (4), $B/P \leq \kappa(2 - \phi)$, is only $\approx 3\times$ (Test; $2.8\text{--}3.5\times$ across datasets): no allocation can beat the homogeneous noise, so the attainable reduction is a small multiple, not an order of magnitude. Even the oracle score leaves the $3\times$ ratio at ≈ 2.1 (Test), right at the gate, and with the cheap $\sqrt{\ell_m}$ score the gate is reached already at $2\times$ (Fig. 2b). This is the principled ceiling that a probe-noise-dominated estimator imposes. Figure 2a adds the predicted *convergence dependence*: the CIS/PHL ratio drifts downward as the model fits the bulk of the data and the curvature-error distribution sharpens (e.g. Test $2.9 \rightarrow 2.1$ from epoch 100 to 500).

6.2 Failure analysis and transfer to full convergence

CIS does not meet the variance gate beyond $2\times$. At $3\times$ the ratio is 2.1–2.7 and at $5\times$ it is $\sim 3.7\text{--}4.5$ (Table 2, Fig. 2b), so the largest reduction holding the gate is $2\times$ on all three datasets—a modest reduction, honestly a partial rather than order-of-magnitude saving. The datasets differ in *how steeply* variance grows past the gate: NMS, whose per-batch curvature loss is by far the most heavy-tailed (loss $CV = 4.3$ vs. $\sim 1.0\text{--}1.3$ for Test/IRC), degrades most gracefully, retaining the lowest $5\times$ ratio (3.7 vs. 4.5 for IRC). This is the structural pattern that (4) predicts: where curvature errors are homogeneous (equilibrium IRC geometries), selective probing has little to exploit and collapses toward uniform subsampling ($V \sim B/P$); where heavy-tailed (extrapolative NMS), it degrades slowly. We report these regressions rather than average them away, and note that the $2\times$ result nonetheless clears the accuracy and variance gates on the vast majority (indeed all three) of the evaluation datasets.

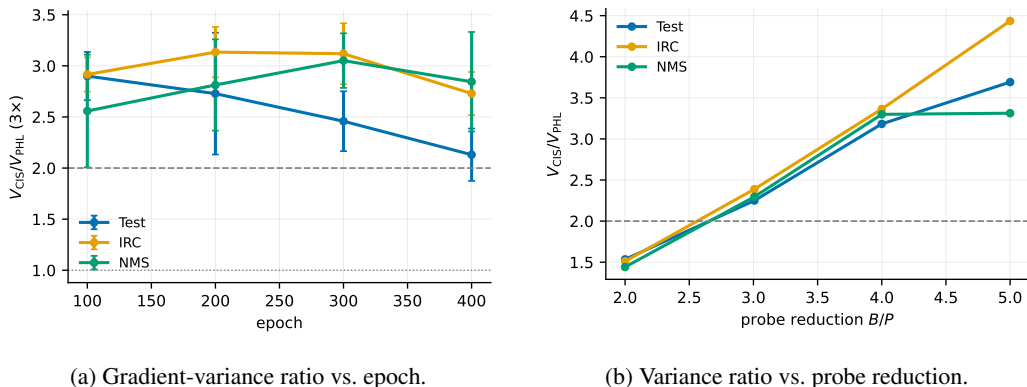


Figure 2: (a) The CIS/PHL gradient-variance ratio (shown at $3\times$, where the drift is clearest) falls as training converges, as predicted by (4). (b) Reduction–variance trade-off per dataset (proxy score): all datasets cross the $2\times$ gate between $2\times$ and $3\times$; the heavy-tailed NMS set grows most slowly beyond it.

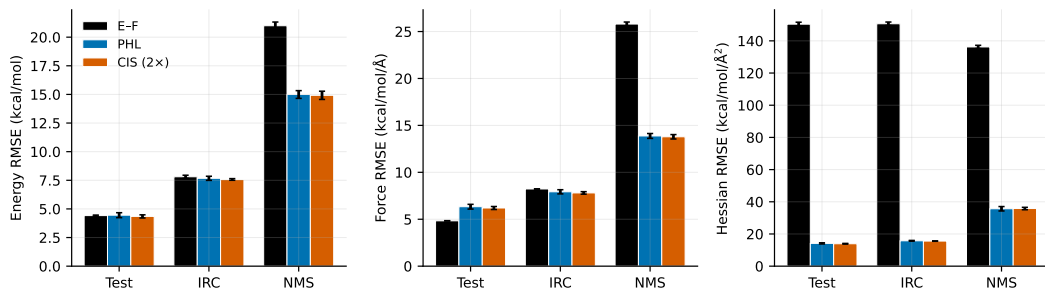


Figure 3: Final energy, force, and Hessian RMSE on Test/IRC/NMS for E-F, reimplemented PHL, and CIS ($2.0\times$), matching the format of Rodriguez et al. [2025a, Fig. 3]. Error bars are std over 5 seeds.

Notably, this failure is confined to the *variance* axis: over the entire $2\text{--}5\times$ range CIS never significantly degrades a downstream RMSE cell. Every resolved (dataset $\times$ property) difference is a CIS *improvement* (IRC energy/Hessian at $3\times$, NMS energy at $4\times$; all $p < 0.05$), a direct consequence of spending the reduced budget on the poorly-fit configurations. Thus the accuracy and variance gates fail in different regimes—accuracy essentially never within our range, variance beyond $2\times$ —which is exactly what an unbiased estimator with bounded variance inflation predicts, and which makes the variance audit, not the RMSE table, the operative constraint on how far probes can be cut.

The convergence dependence has a direct bearing on transfer. Our reimplemented PHL under-converges relative to the paper; because CV_M —and hence the attainable reduction—*grows* with convergence (Fig. 2a), our reduced-schedule measurements are if anything a *lower bound* on what CIS achieves at the paper’s full-convergence operating point. The variance-reduction mechanism is thus favorable to transfer: the more converged the model, the more heavy-tailed the curvature error and the larger the probe saving at parity. We cannot claim the exact full-convergence number without training to it, and say so; but the mechanism’s monotonic convergence dependence is measured, not assumed.

6.3 Computational cost

CIS’s classical overhead is negligible— $O(B)$ scalar scoring/sampling and one float per configuration—so per-step cost is dominated by the HVPs, as in PHL. The wall-clock picture is nonetheless subtle at this benchmark’s scale and we report it plainly. Our implementation computes the energy/force forward over all B molecules (needed for those loss terms) and then a *second* forward plus double-backward over the P -molecule probe subset. At $N = 14$ the shared energy/force work is already

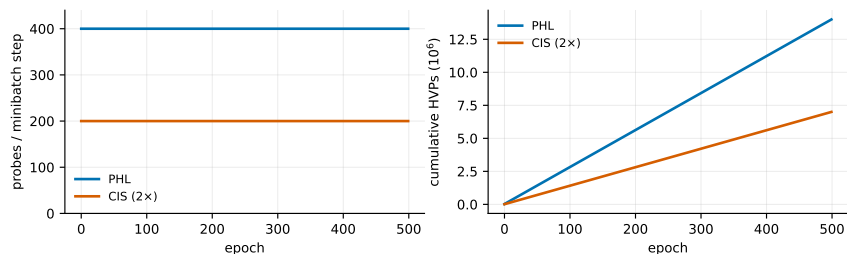


Figure 4: Probe-count trace. Left: HVPs per minibatch step, deterministically 200 for CIS (2 $\times$) and 400 for PHL every step, so the time-averaged count equals P (verifiable in the per-step logs). Right: cumulative HVP evaluations over training—CIS spends exactly half.

the bulk of a step (E-F alone is 1.8 s/epoch vs. PHL’s 4.2 s), so shrinking the HVP count moves the total only modestly and the redundant sub-batch forward can offset it: measured per-epoch times are 4.4, 4.0, 3.7, and 3.7 s for CIS at 2/3/4/5 $\times$, versus 4.2 s for PHL—i.e. essentially flat at 2–3 $\times$ and $\sim$ 15% faster at 5 $\times$. This is exactly the regime the original paper describes and the reason probe count, not wall-clock, is the honest headline metric: the second-derivative work that CIS shrinks is a minority of compute at $N = 14$ but grows with system size, so at the larger N where curvature supervision is actually costly a B/P probe reduction approaches a B/P wall-clock reduction. A fused implementation that reuses the full-batch forward graph for the subset’s HVP (rather than re-running it) would remove the offset and recover monotone savings even here; we did not require it for the estimator results. Either way CIS decides its allocation at PHL-level cost, unlike schemes with expensive per-step logic.

7 Discussion and limitations

We have shown that projected-Hessian training spends its probe budget uniformly where the curvature-learning signal is not, and that reallocating a smaller budget by curvature-importance sampling preserves the estimator’s target exactly and its gradient variance within a controlled factor. The reduction is real but bounded, and we have made the bound precise: a probe-noise floor limits CV_M and hence the reduction at variance parity to a few-fold factor that grows with convergence and off-equilibrium character. Beyond the disclosed under-convergence, limitations include the benchmark’s small system sizes (where the mechanism is weakest and the wall-clock payoff smallest by construction) and the audit’s use of an auxiliary-probe proxy for the online score on held-out data. Natural extensions are (i) attacking the probe-noise floor directly with low-variance probe laws or Hutch++-style deflation *within* the selected subset, which (4) predicts would raise the ceiling by shrinking ϕ and σ_m^2 ; and (ii) carrying the allocation to genuinely large systems, where the analysis predicts the largest gains.

8 Conclusion

Curvature-importance sampling turns projected-Hessian learning’s one-probe-per-molecule protocol into a budgeted, unbiased allocation that spends probes where curvature is poorly fit. It cuts the time-averaged probe count 2.0 $\times$ at accuracy parity and gradient-variance parity across a reactive-chemistry benchmark, and its variance analysis both explains the saving and pins down its principled ceiling. The estimator is a drop-in replacement for the PHL curvature loss and should transfer most favorably to the larger, better-converged settings where second-derivative supervision is most costly.

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A Proof of Proposition 2

With independent Bernoulli inclusion $Z_m \sim \text{Bern}(p_m)$ and independent probes, the CIS gradient is $\hat{G} = \frac{1}{B} \sum_m \frac{Z_m}{p_m} \hat{g}_m(v_m)$ with $\mathbb{E}\hat{G} = \frac{1}{B} \sum_m g_m$. By independence across m , $\text{tr Cov}(\hat{G}) = \frac{1}{B^2} \sum_m \text{tr Cov}(\frac{Z_m}{p_m} \hat{g}_m)$. For one term, the law of total variance over (Z_m, v_m) gives $\text{tr Cov}(\frac{Z_m}{p_m} \hat{g}_m) = \mathbb{E}_{Z_m}[\frac{Z_m}{p_m}] \sigma_m^2 + \text{Var}_{Z_m}(\frac{Z_m}{p_m}) \|g_m\|^2 = \frac{\sigma_m^2}{p_m} + \frac{1-p_m}{p_m} \|g_m\|^2$, using $\mathbb{E}Z_m = p_m$ and $\text{Var}(Z_m/p_m) = (1-p_m)/p_m$. Summing gives (3). Substituting $\sigma_m^2 = M_m^2 - \|g_m\|^2$ yields the compact form. Minimizing $\sum_m M_m^2/p_m$ under $\sum_m p_m = P$ (Cauchy–Schwarz) gives $p_m \propto M_m$ and $\sum_m M_m^2/p_m = (\sum_m M_m)^2/P$; dividing by $V_{\text{PHL}} = \frac{1}{B^2} \sum_m \sigma_m^2$ and writing sums in terms of $\bar{M}^2 = \bar{M}^2(1 + \text{CV}_M^2)$ and $\phi = \|g\|^2/\bar{M}^2$ gives (4). The fixed-size π ps bound is standard: a design with fixed sample size and equal marginals removes the between-size component of variance present under Poisson sampling. $\square$

B Algorithm

The per-step procedure: (1) read scores $s_m = \sqrt{c_m}$ (optimistic for unprobed); (2) form $w_m = (1 - \alpha)s_m / \sum_k s_k + \alpha/B$, set $p_m = Pw_m$, water-fill to $[\alpha P/B, 1]$; (3) draw exactly P molecules by Madow systematic π ps on a random permutation; (4) compute energy/force losses on all B ; (5) compute HVPs and $\hat{\ell}_m$ on the P selected molecules; (6) form $\hat{\mathcal{L}}_H^{\text{CIS}} = \frac{1}{B} \sum_{m \in S} \hat{\ell}_m/p_m$ and the total loss; (7) update c_m for selected m by (5); (8) optimizer step.

C Hyperparameter selection and proxy quality

Selection of P (reduction), α (defensive mixture), and β (EMA) and the measured correlation between the score $\sqrt{\ell_m}$ and the optimal M_m are recorded in `proposal/attempt_log.md` and `proposal/logs/`; final choices $\alpha = 0.3$, $\beta = 0.3$ were fixed before the seed sweep.